Bill Of Materials: 3D Positioning

Part Name/Description	Unit Quantit	Unit Prototype Co	Total Prototype Co	Unit Production Co	Total Production Co	Manufacturer	Manufacturer Part #	Vendor Link	Datasheet Link	Supplier	Supplier Part	# Ordere	Date Ordere	# Receive	Surplus	Schematic Reference Designators
Inductor 3.3uH	1	$3.15	$3.15	$3.15	$3.15	Coilcraft	XAL4030-332MEC	DigiKey	Datasheet Link	DigiKey	2457-XAL4030	3	3/6/2026	0	-1	L
100nf (0.1 µF) Ceramic Capacitor, +/-10%, X7R, 50V, 080	4	$0.10	$0.40	$0.10	$0.40	Samsung Electro-Me	CL21F104ZAANNNC	DigiKey	Datasheet Link	DigiKey	1276-1286-2-N	5	3/6/2026	0	-4	C1,C2,C3,C4
0.1 µF Ceramic Capacitor, +/-10%, X7R, 50V, 0805 packa	5	$0.10	$0.50	$0.10	$0.50	Samsung Electro-Me	CL21F104ZAANNNC	DigiKey	Datasheet Link	DigiKey	1276-1286-2-N	5	3/6/2026	0	-5	C5,C6C15
100uF 50 V Aluminum Electrolytic Capacitor	1	$0.75	$0.75	$0.75	$0.75	Panasonic Electronic	EEE-HA1H101P	DigiKey	Datasheet Link	DigiKey	PCE4210TR-N	3	3/6/2026	0	-1	C7
330uF 10 V Aluminum Electrolytic Capacitor	1	$0.66	$0.66	$0.66	$0.66	Panasonic Electronic	EEE-1AA331P	DigiKey	Datasheet Link	DigiKey	PCE3874TR-N	3	3/6/2026	0	-1	C8
10uF 10V Ceramic Capacitors	3	$0.08	$0.24	$0.08	$0.24	Taiyo Yuden	LMK212BJ106KG-T	DigiKey	Datasheet Link	DigiKey	587-1300-2-ND	5	3/6/2026	0	-3	C9,C10,C11
22uF 4 Volts Ceramic Capacitor	3	$0.12	$0.36	$0.12	$0.36	Murata Electronics	GRM188R60G226MEA	DigiKey	Datasheet Link	DigiKey	490-5526-2-ND	6	3/6/2026	0	-3	C12,C13,C14
Green LEDs, LTST-C150GKT	4	$0.15	$0.60	$0.15	$0.60	Lite-On Inc.	LTST-C150GKT	DigiKey	Datasheet Link	DigiKey	160-1169-2-ND	8	3/6/2026	0	-4	D1,D2,D3,D4
Schotty Diode	2	$0.10	$0.20	$0.10	$0.20	Vishay General Semi	MSS1P4-M3/89A	DigiKey	Datasheet Link	DigiKey	MSS1P4-M3/8	4	3/6/2026	0	-2	D5,D6
Fuse Holder	2	$1.25	$2.50	$1.25	$2.50	Keystone Electronics	3588	DigiKey	Datasheet Link	DigiKey	36-3588-ND	4	3/6/2026	0	-2	F1
Servos (1142)	2	$19.95	$39.90	$19.95	$39.90	Adafruit Industries LL	1142	DigiKey	Datasheet Link	DigiKey	1528-1083-ND	4	3/6/2026	0	-2	J1,J2
Conn_02x04_Pin	2	$0.00	$0.00	$0.00	$0.00	PRLTA 110	PRLTA 109	PRLTA 109	Datasheet Link	PRLTA 109	PRLTA 109	4	3/6/2026	0	-2	J3,J4
Inductor 220uH	1	$0.90	$0.90	$0.90	$0.90	Eaton - Electronics D	DR124-221-R	DigiKey	Datasheet Link	DigiKey	283-DR124-22	3	3/6/2026	0	-1	L1
4.75 kOhms 0.125W, 1/8W Resistor 0805	8	$0.00	$0.00	$0.00	$0.00	PRLTA 109	PRLTA 109	DigiKey	Datasheet Link	PRLTA 109	PRLTA 109	10	3/6/2026	0	-8	R1,R2,R3,R4,R5,R6,R7,R8
330 Ohms 0.1W 1/10w Resistor 0805	4	$0.10	$0.40	$0.10	$0.40	PRLTA 109	PRLTA 109	DigiKey	Datasheet Link	PRLTA 109	PRLTA 109	10	3/6/2026	0	-4	R9,R10,R11,R12
10 kOhm 1/10W Resistor 0805	2	$0.10	$0.20	$0.10	$0.20	PRLTA 109	PRLTA 109	DigiKey	Datasheet Link	PRLTA 109	PRLTA 109	10	3/6/2026	0	-2	R13,R14
30 kOhms 1/8W Resistor 0805	2	$0.10	$0.20	$0.20	$0.40	PRLTA 109	PRLTA 109	DigiKey	Datasheet Link	PRLTA 109	PRLTA 109	10	3/6/2026	0	-2	R15,16
20 kOhms 1/16W Resistor 0805	1	$0.10	$0.10	$0.20	$0.20	PRLTA 109	PRLTA 109	DigiKey	Datasheet Link	PRLTA 109	PRLTA 109	10	3/6/2026	0	-1	R17
220 kOhms 1/16W Resistor 0805	1	$0.10	$0.10	$0.10	$0.10	PRLTA 109	PRLTA 109	DigiKey	Datasheet Link	PRLTA 109	PRLTA 109	10	3/6/2026	0	-1	R18
Switch Tactile 0.05A 12V	2	$0.27	$0.54	$0.27	$0.54	PRLTA 109	PRLTA 109	DigiKey	Datasheet Link	PRLTA 109	PRLTA 109	4	3/6/2026	0	-2	SW1,SW2
AS5600L Hall Effect	2	$3.20	$6.40	$3.20	$6.40	ams-OSRAM USA IN	AS5600L-ASOM SOIC8	DigiKey	Datasheet Link	PRLTA 109	PRLTA 109	3	3/6/2026	0	-2	U1,U3
AS5600 Hall Effect	2	$3.17	$6.34	$3.17	$6.34	ams-OSRAM USA IN	AS5600-ASOM SOIC8 L	DigiKey	Datasheet Link	PRLTA 109	PRLTA 109	3	3/6/2026	0	-2	U2,U4
Esp32-S3-Wroom-N4	1	$5.06	$5.06	$5.06	$5.06	Espressif Systems	ESP32-S3-WROOM-1-N	Digikey	Datasheet Link	PRLTA 109	PRLTA 109	2	3/6/2026	0	-1	U5
TPS56524DRLR	1	$0.65	$0.65	$0.65	$0.65	Texas Instruments	TPS565242DRLR	DigiKey	Datasheet Link	DigiKey	296-TPS56524	3	3/6/2026	0	-1	U6
LM2575T-3.3G	1	$2.51	$2.51	$2.51	$2.51	onsemi	LM2575T-3.3G	DigiKey	Datasheet Link	PRLTA 109	PRLTA 109	3	3/6/2026	0	-1	U7
Barrel Jack Switch	1	$0.69	$0.69	$0.69	$0.69	Kycon, Inc.	KLDX-0202-A	DigiKey	Datasheet Link	PRLTA 109	2092-KLDX-02	2	3/6/2026	0	-1	U8
Micro-B USB Connector	1	$0.78	$0.78	$0.78	$0.78	GCT	USB3131-30-0230-A	DigiKey	Datasheet Link	PRLTA 109	PRLTA 109	3	3/6/2026	0	-1	U9
Conn_0x02_Pin	1	$0.00	$0.00	$0.00	$0.00	PRLTA 109	PRLTA 109	Digikey	Datasheet Link	PRLTA 109	PRLTA 109	4	3/11/2026	0	-1	JP1
Test_Points	31	$0.13	$4.03	$0.13	$4.03	KOA Speer Electronic	RCWCTE	Digikey	Datasheet Link	DigiKey	2019-RCWCTE	35	3/11/2026	0	-31	TP 1-31